

A. Additional Experimental Results

We additionally evaluate JAMB on the Dominant-select task category from the RoboTwin 2.0 benchmark. Table VIII complements our main bimanual evaluation by demonstrating JAMB’s effectiveness on tasks that require selecting the appropriate arm.

B. Future Track Prediction Metrics

We evaluate future 3D track prediction using average displacement error (ADE) and final displacement error (FDE). Both metrics are computed over moving patches, defined as queries whose ground-truth endpoint displacement satisfies $\|P_t(i, H_p)\|_2 > 0.01$ m. For each validation sample, we re-index these queries as $i = 1, \dots, N_{\text{mov}}$ and define

$$\text{ADE} = \frac{1}{N_{\text{mov}} H_p} \sum_{i=1}^{N_{\text{mov}}} \sum_{h=1}^{H_p} \left\| \hat{P}_t(i, h) - P_t(i, h) \right\|_2, \quad (17)$$

$$\text{FDE} = \frac{1}{N_{\text{mov}}} \sum_{i=1}^{N_{\text{mov}}} \left\| \hat{P}_t(i, H_p) - P_t(i, H_p) \right\|_2, \quad (18)$$

where N_{mov} is the number of selected moving queries, H_p is the prediction horizon, and $\hat{P}_t(i, h)$ and $P_t(i, h)$ denote the predicted and ground-truth 3D displacements relative to the query’s anchor position, respectively. ADE averages the Euclidean displacement error across the full prediction horizon, whereas FDE measures the error at the final step. Both metrics use de-normalized displacements in meters, converted to millimeters for reporting. Lower values indicate better prediction accuracy.

C. Ablation Settings and Per-Task Results

We construct ablation variants from the full model to examine future-track supervision, action-track coupling, visual-track fusion, and spatiotemporal grounding. Unless otherwise specified, we retain the remaining model components and training settings.

a) Ablation settings: The variants are implemented as follows:

- **Action-only.** This variant retains the action-generation backbone but removes the track-prediction branch and all 3D track supervision.
- **Track regression.** We replace track diffusion with deterministic regression from fixed track queries. Each query remains concatenated with its corresponding visual feature and participates in shared self-attention with action tokens. The regression head predicts clean tracks at every action-denoising step, but these predictions are not fed back as an evolving track diffusion state. We retain the prediction from the final denoiser forward pass.
- **Track-conditioned action diffusion.** This variant uses two-stage prediction. In the first stage, noisy 3D track tokens are concatenated with their corresponding patch-level visual features and denoised to predict future 3D tracks. In the second stage, the predicted tracks are

concatenated with the corresponding visual features and participate in self-attention with action tokens during action denoising. The track predictions remain fixed throughout the second stage.

- **w/o Visual-Track Concatenation.** We remove the concatenation of each track representation with its corresponding patch-level visual feature. Instead, track and action tokens interact with separate visual tokens through cross-attention. The model must therefore learn visual-track associations through attention rather than receiving explicitly paired features.
- **w/o 4D RoPE.** We remove the rotary encoding of spatial coordinates and trajectory time while retaining learnable positional embeddings for token identity and sequence role. All other components of joint action-track denoising remain unchanged.

b) Evaluation protocol: Table IX reports success rates on the eight RoboTwin tasks used for ablation. For each variant and task, we evaluate the trained policy using three evaluation seeds, with 50 rollouts per seed, and report the mean success rate across seeds. Table X reports average displacement error (ADE) and final displacement error (FDE) for moving patches on the corresponding validation sets. Both metrics are computed using de-normalized 3D displacements, reported in millimeters, and averaged across tasks.

c) Track prediction and policy performance: The full model achieves the lowest ADE/FDE of 7.6/10.0 mm and the highest average policy success rate of 81.7%. Compared with track-conditioned action diffusion, the full model reduces ADE/FDE from 9.9/12.5 mm and improves success by 5.4 percentage points. Removing 4D RoPE increases ADE/FDE to 13.2/17.2 mm and reduces success to 74.1%, supporting the benefits of shared spatiotemporal grounding for both prediction and control. Removing visual-track concatenation also increases ADE/FDE to 11.9/15.3 mm and reduces success to 70.0%. This comparison supports the effectiveness of the full model’s patch-aligned fusion design relative to the separate-token cross-attention variant.

However, lower track error does not consistently translate into higher policy success. Track regression predicts more accurate tracks than the variant without 4D RoPE (11.1/14.8 mm vs. 13.2/17.2 mm), yet their success rates are similar (73.9% vs. 74.1%). Likewise, the variant without visual-track concatenation has lower track errors than the variant without 4D RoPE, but a lower success rate (70.0% vs. 74.1%). These results suggest that track accuracy alone does not explain policy performance; the way motion representations interact with visual features and action generation also matters. The full model combines accurate future-track prediction with effective action generation, although these comparisons do not isolate a causal effect of track accuracy on manipulation success.

D. Per-Task Easy-to-Hard Generalization Results

Table XI reports per-task results when transferring policies trained on Easy scenes to Hard scenes with unseen textures and clutter, without fine-tuning. Each policy is evaluated

TABLE VIII

COMPARISON ON DOMINANT-SELECT TASKS (16 TASKS). TASKS REQUIRING APPROPRIATE ARM SELECTION. WE REPORT TASK SUCCESS RATE (%) AS MEAN $\pm$ STANDARD DEVIATION ACROSS THREE EVALUATION SEEDS. BEST RESULTS ARE SHOWN IN **BOLD** AND SECOND-BEST RESULTS ARE UNDERLINED.

Method	Avg. $\uparrow$ (%)	Beat Block Hammer	Place Shoe	Move Can Pot	Open Laptop	Open Microwave	Place A2B Left	Place A2B Right
DP [10]	41.7	92.7 ± 5.0	46.0 ± 7.1	70.0 ± 15.6	64.0 ± 4.3	0.0 ± 0.0	17.3 ± 4.7	17.3 ± 2.5
DP3 [17]	64.4	<u>88.0</u> ± 1.6	62.0 ± 2.8	92.7 ± 1.9	39.3 ± 1.9	22.0 ± 1.6	53.3 ± 0.9	50.7 ± 5.7
DICP [8]	48.3	86.0 ± 3.3	<u>72.0</u> ± 7.1	74.7 ± 3.4	80.0 ± 4.3	16.0 ± 4.3	8.0 ± 1.6	10.0 ± 1.6
GAP [9]	51.7	39.3 ± 1.9	60.7 ± 2.5	66.0 ± 8.8	<u>81.3</u> ± 2.5	<u>36.0</u> ± 8.0	14.0 ± 1.6	11.3 ± 3.4
ATM [32]	47.5	84.7 ± 2.5	40.7 ± 5.7	<u>91.3</u> ± 3.8	68.7 ± 3.8	1.3 ± 1.9	<u>24.7</u> ± 2.5	14.7 ± 0.9
JAMB	73.5	86.7 ± 6.2	96.0 ± 1.6	90.0 ± 2.8	92.7 ± 0.9	83.3 ± 2.5	23.3 ± 4.1	<u>24.0</u> ± 2.8
Turn Switch	Place Fan	Place Container Plate	Press Stapler	Place Phone Stand	Rotate QR Code	Place Empty Cup	Place Object Stand	Shake Bottle
17.3 ± 3.4	4.7 ± 4.1	51.3 ± 3.4	20.0 ± 1.6	38.0 ± 4.9	24.0 ± 1.6	61.3 ± 4.1	48.7 ± 1.9	94.7 ± 1.9
<u>36.7</u> ± 6.6	<u>50.7</u> ± 5.0	<u>92.0</u> ± 1.6	67.3 ± 2.5	54.0 ± 4.9	69.3 ± 2.5	<u>75.3</u> ± 8.2	76.7 ± 3.4	100.0 ± 0.0
25.3 ± 3.8	40.0 ± 4.3	86.7 ± 4.7	39.3 ± 0.9	11.3 ± 5.7	50.7 ± 1.9	52.0 ± 3.3	36.0 ± 4.3	84.0 ± 1.6
34.7 ± 1.9	45.3 ± 4.7	78.0 ± 9.1	68.0 ± 3.3	30.0 ± 8.6	62.0 ± 4.3	61.3 ± 6.6	39.3 ± 4.1	<u>99.3</u> ± 0.9
23.3 ± 2.5	4.0 ± 0.0	69.3 ± 8.1	31.3 ± 6.2	<u>58.0</u> ± 4.9	48.0 ± 4.3	64.7 ± 8.1	40.7 ± 6.6	94.0 ± 4.3
38.7 ± 1.9	62.7 ± 1.9	98.0 ± 1.6	80.7 ± 7.7	90.0 ± 2.8	<u>62.0</u> ± 2.8	94.7 ± 0.9	<u>54.7</u> ± 8.4	98.7 ± 0.9

TABLE IX

PER-TASK ABLATION RESULTS ON EIGHT ROBOTWIN TASKS. WE REPORT PER-TASK AVERAGE SUCCESS RATE (%) ACROSS THREE EVALUATION SEEDS, WITH 50 ROLLOUTS PER SEED PER TASK. BEST RESULTS ARE SHOWN IN **BOLD** AND SECOND-BEST RESULTS ARE UNDERLINED.

Variant	Avg. $\uparrow$ (%)	Handover Block	Stack Blocks Two	Stack Blocks Three	Scan Object	Place Bread Skillet	Place Dual Shoes	Pick Diverse Bottles	Place Bread Basket
Action-only	61.4	67.3	71.3	59.3	63.3	48.0	43.3	74.0	64.7
Track regression	73.9	94.7	83.3	80.0	59.3	70.0	60.7	77.0	66.0
Track-conditioned action diffusion	<u>76.3</u>	<u>95.0</u>	85.3	84.0	54.7	75.0	59.0	87.0	70.0
w/o Visual-Track Concatenation	70.0	81.3	78.0	70.0	60.0	68.7	67.3	74.7	60.0
w/o 4D RoPE	74.1	86.7	<u>90.7</u>	<u>86.7</u>	<u>66.0</u>	67.3	<u>62.0</u>	61.3	<u>72.0</u>
Full model	81.7	96.7	91.3	96.0	72.0	<u>72.7</u>	<u>62.0</u>	<u>85.3</u>	77.3

TABLE X

FUTURE 3D TRACK PREDICTION ON THE VALIDATION SETS OF THE EIGHT ROBOTWIN TASKS. ADE AND FDE ARE COMPUTED OVER MOVING PATCHES AND AVERAGED ACROSS TASKS. BEST RESULTS ARE SHOWN IN **BOLD**.

Variant	ADE (mm) $\downarrow$	FDE (mm) $\downarrow$
Track regression	11.1	14.8
Track-conditioned action diffusion	9.9	12.5
w/o Visual-Track Concatenation	11.9	15.3
w/o 4D RoPE	13.2	17.2
Full model	7.6	10.0

using three evaluation seeds with 50 rollouts per seed, and we report the mean and standard deviation across these seeds. JAMB achieves the highest success rate on seven of the eight tasks, with an average of 17.9% compared with 4.3% for the strongest baseline, GAP. However, success rates remain low overall, and JAMB does not succeed on Place Bread Skillet, highlighting the remaining challenges of generalization under these visual changes.

E. Additional Future Prediction Targets

We further examine whether predicting richer future observations provides additional benefits beyond 3D tracks. Starting from the full JAMB model, we additionally predict either future RGB observations or future 3D geometry features,

while retaining the same DiT backbone configuration and evaluation protocol.

For future-image prediction, we encode RGB observations using a frozen variational autoencoder (VAE) from Stable Diffusion XL (SDXL) and supervise the predicted VAE latents. For endpoint geometry prediction, we extract target features using a frozen pretrained Pi3 encoder and apply supervision directly in feature space. Although these features can be decoded into a point map, the loss is computed on the features rather than the decoded geometry. In both variants, the additional latent or feature tokens are incorporated into the DiT sequence, grounded by 4D RoPE, and jointly denoised with actions and tracks.

Table XII compares these variants on the same eight RoboTwin tasks used for ablation. Adding future-image or geometry-feature prediction reduces average success from 81.7% to 79.4% and 74.2%, respectively. Under the evaluated configuration, these additional prediction targets provide no further performance benefit over joint action-track prediction alone.

F. Alternative Future Representations

We compare different future prediction targets within our framework on the same eight RoboTwin tasks. The full-horizon 2D track variant replaces 3D displacement targets with 2D pixel displacements over the same prediction horizon, with the prediction head adjusted accordingly. We

TABLE XI

PER-TASK EASY-TO-HARD GENERALIZATION ON EIGHT ROBOTWIN TASKS. ALL METHODS ARE TRAINED ON *Easy* AND EVALUATED ON *Hard* WITHOUT FINE-TUNING. WE REPORT TASK SUCCESS RATE (%) AS MEAN $\pm$ STANDARD DEVIATION ACROSS THREE EVALUATION SEEDS. BEST RESULTS ARE SHOWN IN **BOLD** AND SECOND-BEST RESULTS ARE UNDERLINED.

Method	Avg. $\uparrow$ (%)	Handover Block	Stack Blocks Two	Stack Blocks Three	Scan Object	Place Bread Skillet	Place Dual Shoes	Pick Diverse Bottles	Place Bread Basket
DP3 [17]	1.7	4.7 $\pm$ 2.5	<u>1.3</u> $\pm$ 0.9	0.0 $\pm$ 0.0	2.0 $\pm$ 1.6	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	2.7 $\pm$ 1.9	2.7 $\pm$ 2.5
DICP [8]	3.9	0.7 $\pm$ 0.9	0.7 $\pm$ 0.9	0.0 $\pm$ 0.0	<u>4.7</u> $\pm$ 2.5	8.7 $\pm$ 3.4	3.3 $\pm$ 2.5	4.7 $\pm$ 3.4	<u>8.0</u> $\pm$ 6.5
GAP [9]	4.3	<u>6.0</u> $\pm$ 1.6	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	3.3 $\pm$ 2.5	<u>2.0</u> $\pm$ 2.8	<u>8.0</u> $\pm$ 1.6	<u>9.3</u> $\pm$ 1.9	6.0 $\pm$ 1.6
ATM [32]	0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
JAMB	17.9	12.0 $\pm$ 1.6	10.7 $\pm$ 3.8	22.0 $\pm$ 3.3	13.3 $\pm$ 2.5	0.0 $\pm$ 0.0	12.7 $\pm$ 3.4	47.3 $\pm$ 6.6	25.3 $\pm$ 5.2

TABLE XII

EFFECT OF ADDITIONAL FUTURE PREDICTION TARGETS ON EIGHT ROBOTWIN TASKS. WE REPORT AVERAGE SUCCESS RATES (%) OVER THREE EVALUATION SEEDS, WITH 50 ROLLOUTS PER SEED PER TASK. BEST RESULTS ARE SHOWN IN **BOLD**.

Future prediction target	Avg. $\uparrow$	Sync.	Seq.
3D tracks	81.7	74.3	89.0
3D tracks + future image	79.4	71.8	87.0
3D tracks + future 3D geometry	74.2	64.5	83.9

evaluate this variant both with and without 4D RoPE. When enabled, 4D RoPE retains the same depth-derived world-space anchors as the full model, even though the prediction targets are in image space.

The endpoint 3D displacement variant predicts only the displacement at the final horizon step, whereas the full model predicts displacements at all future steps. Both variants retain patch-aligned visual-track fusion. For the endpoint geometry variant, we replace the track prediction target with Pi3 features extracted by a frozen pretrained encoder. Patch-level noisy geometry features are concatenated with their corresponding visual features, and supervision is applied directly in feature space rather than to the decoded point map. All variants jointly denoise their respective future representations with actions.

As shown in Table XIII, adding 4D RoPE to full-horizon 2D tracks improves average success from 76.1% to 79.5%, supporting the value of spatiotemporal grounding across modalities even when motion is predicted in image space. Endpoint 3D displacements and full-horizon 3D tracks achieve comparable success rates of 81.2% and 81.7%, respectively, while endpoint 3D geometry achieves 82.8%. Performance across these three 3D prediction targets falls within 1.6 percentage points, demonstrating that our joint prediction architecture accommodates different future representations with comparable manipulation success. Our track-based formulation achieves comparable performance using depth-derived spatial anchors, without the additional Pi3 feature extraction used by the geometry variant.

We further compare these representations under an Easy-to-Hard visual shift, where all variants are trained on Easy scenes and evaluated on Hard scenes without fine-tuning. Despite their similar in-distribution performance, full-horizon

TABLE XIII

COMPARISON OF ALTERNATIVE FUTURE REPRESENTATIONS ON EIGHT ROBOTWIN TASKS. WE REPORT AVERAGE SUCCESS RATES (%) OVER THREE EVALUATION SEEDS, WITH 50 ROLLOUTS PER SEED PER TASK. BEST RESULTS ARE SHOWN IN **BOLD** AND SECOND-BEST RESULTS ARE UNDERLINED.

Future prediction target	Avg. $\uparrow$	Sync.	Seq.
2D tracks (full horizon)	76.1	67.5	84.7
2D tracks + 4D RoPE (full horizon)	79.5	70.2	88.8
3D geometry (endpoint)	82.8	76.7	88.9
3D point displacements (endpoint)	81.2	73.0	89.4
3D tracks (full horizon, ours)	<u>81.7</u>	<u>74.3</u>	<u>89.0</u>

3D tracks achieve higher Easy-to-Hard success (17.9%) than endpoint displacements (12.2%) and endpoint geometry (10.3%), as shown in Table XIV. These results suggest that full-horizon 3D tracks support better generalization under the evaluated visual shift. One possible explanation is that endpoint geometry features encode surrounding scene structure and appearance, potentially making them more sensitive to changes in background and clutter. Endpoint displacements focus on point motion but supervise only the net change at the final step. Full-horizon tracks additionally supervise intermediate motion, which may encourage the model to learn task-relevant motion patterns that transfer across visual conditions. These explanations remain hypotheses rather than mechanisms isolated by the current experiments.

G. Additional Implementation Details

1) *Architecture and Optimization:* For real-world experiments, the policy uses a single head-camera RGB view with an observation horizon of $H_o = 1$. We crop regions outside the tabletop workspace, resulting in input images of 168×322 pixels (height $\times$ width). DINOv2 [44] encodes these images with a patch size of 14, yielding a 12×23 patch grid and $N = 276$ track queries, one per visual patch. Both the action and track prediction horizons are set to $H_a = H_p = 16$.

JAMB uses a DiT with 4 Transformer blocks, a hidden dimension of 1024, and 8 attention heads. We optimize the model with AdamW using a learning rate of 10^{-4} , $(\beta_1, \beta_2) = (0.9, 0.999)$, weight decay 10^{-6} , and a batch size of 64. The learning-rate schedule consists of 500 warm-up steps followed by cosine decay.

TABLE XIV

PER-TASK EASY-TO-HARD GENERALIZATION WITH ALTERNATIVE FUTURE REPRESENTATIONS ON EIGHT ROBOTWIN TASKS. WE REPORT TASK SUCCESS RATE (%) AS MEAN $\pm$ STANDARD DEVIATION ACROSS THREE EVALUATION SEEDS. BEST RESULTS ARE SHOWN IN **BOLD**.

Future prediction target	Avg. $\uparrow$ (%)	Handover Block	Stack Blocks Two	Stack Blocks Three	Scan Object	Place Bread Skillet	Place Dual Shoes	Pick Diverse Bottles	Place Bread Basket
2D tracks (full horizon)	9.3	2.7 $\pm$ 2.5	16.7 $\pm$ 3.4	22.0 $\pm$ 4.3	3.3 $\pm$ 0.9	0.0 $\pm$ 0.0	5.3 $\pm$ 2.5	16.0 $\pm$ 1.6	8.7 $\pm$ 2.5
2D tracks + 4D RoPE (full horizon)	8.8	0.0 $\pm$ 0.0	12.7 $\pm$ 3.4	3.3 $\pm$ 3.4	8.7 $\pm$ 5.2	0.0 $\pm$ 0.0	6.7 $\pm$ 1.9	31.3 $\pm$ 8.2	7.3 $\pm$ 3.4
3D geometry (endpoint)	10.3	2.7 $\pm$ 0.9	11.3 $\pm$ 5.0	1.3 $\pm$ 1.9	10.7 $\pm$ 2.5	0.0 $\pm$ 0.0	10.7 $\pm$ 4.7	28.7 $\pm$ 5.2	17.3 $\pm$ 5.2
3D point displacements (endpoint)	12.2	7.3 $\pm$ 1.9	26.7 $\pm$ 8.2	5.3 $\pm$ 1.9	13.3 $\pm$ 3.8	0.0 $\pm$ 0.0	2.7 $\pm$ 0.9	31.3 $\pm$ 2.5	11.3 $\pm$ 2.5
3D tracks (full horizon, ours)	17.9	12.0 $\pm$ 1.6	10.7 $\pm$ 3.8	22.0 $\pm$ 3.3	13.3 $\pm$ 2.5	0.0 $\pm$ 0.0	12.7 $\pm$ 3.4	47.3 $\pm$ 6.6	25.3 $\pm$ 5.2

2) *Diffusion and Loss Hyperparameters*: We use clean-sample prediction (prediction_type=sample) for both action and track diffusion, supervising the outputs $\hat{Y}_A$ and $\hat{Y}_P$ with normalized clean targets:

$$Y_A = \tilde{A}_t^0, \quad Y_P = \tilde{P}_t^0. \quad (19)$$

Both branches use the same cosine noise schedule (squaredcos_cap_v2) with $K = 100$ training diffusion timesteps and 10 DDIM denoising steps at inference.

We set $\lambda_{\text{track}} = 10^{-4}$, $w_{\min} = 0$, and $\tau = 0.01$ m, with sigmoid steepness $\kappa = 5/\tau = 500 \text{ m}^{-1}$. The track loss sums squared errors over the prediction horizon and three spatial coordinates, weighted by $w_{b,i}$ for query i in batch element b , and divides by $3 \max(\sum_{b,i} w_{b,i}, 10^{-6})$. It is therefore not averaged over the prediction horizon.

3) *Track Validity and Invalid-Query Supervision*: For simulator-generated track supervision, a patch query is valid if its center has a positive, finite depth value sampled by bilinear interpolation, and its segmentation ID belongs to a tracked rigid body. Tracked bodies include objects and robot arm links. Queries with invalid depth or segmentation IDs outside this set are assigned zero displacement targets.

For each valid query, we transform its anchor-frame 3D position into the owning body’s local frame and compute future positions using that body’s ground-truth poses. Validity is determined at the anchor frame and propagated across the prediction horizon; future visibility and occlusion are not re-evaluated. For prediction steps beyond the episode boundary, we reuse the final recorded pose.

We normalize 3D track displacements as $\tilde{P}_t^0 = P_t^0 / \sigma_{\text{rms}}$, where σ_{rms} is the root mean square of all valid displacement components across samples and prediction steps in each training dataset. This scalar scale is shared across the three spatial dimensions, with no mean subtraction, so zero displacement targets remain zero after normalization. Invalid queries are excluded from scale estimation but included in the movement-weighted track loss without validity masking. Under the specified loss hyperparameters, they receive a weight of $\text{sigmoid}(-5) \approx 0.0067$ and therefore contribute weak zero-displacement supervision.

4) *Training Duration and Checkpoint Selection*: DP and DP3 are trained for 600, 3000 epochs, respectively. JAMB, DICP, and GAP are each trained for 200 epochs. For ATM, the Track Transformer is trained for 100 epochs, while the ATM Diffusion Policy is trained for 200 epochs. For every

method, we use the checkpoint from the final training epoch for evaluation.

5) *4D-RoPE Implementation*: We apply four-dimensional rotary positional encoding (4D RoPE) to incorporate relative spatial and temporal relationships into multimodal attention. Each grounded token is associated with a coordinate $\mathbf{c} = (x, y, z, u)$, where (x, y, z) denotes its position in a common world frame, measured in meters, and u denotes its physical timestep index rather than the diffusion timestep.

a) *Axis allocation and coordinate scaling*: Each attention head has dimension 128, which we divide into four groups of 32 dimensions for the x , y , z , and temporal axes. RoPE is applied independently to each group of the query and key vectors. For an axis coordinate c , the rotation angle for the j -th pair of feature dimensions is

$$\phi_j(c) = \frac{c}{s} b^{-2j/d}, \quad j = 0, \dots, d/2 - 1, \quad (20)$$

where $d = 32$ is the dimension allocated to each axis. We use a frequency base of $b = 100$ for all four axes, a spatial scale of $s_{xyz} = 0.01$ m, and a temporal scale of $s_u = 16$. These rotations encode relative coordinate offsets in query–key dot products.

b) *Token coordinates*: Visual patches and their associated track queries are anchored at the corresponding current 3D patch centers and assigned temporal index $u = 0$. The fused visual–track tokens inherit these coordinates. Each arm-state token is anchored at that arm’s current end-effector position, also with $u = 0$. Action tokens are assigned temporal indices $u = h$ for $h = 1, \dots, H_a$ and spatial coordinates computed from the current noisy action sequence, as described below. Special tokens, including the CLS token when present and the diffusion-timestep token, are excluded from rotary encoding.

Each track token represents the complete future trajectory of one query point rather than an individual future timestep. Its temporal coordinate therefore identifies the current reference frame from which the trajectory is predicted. The ordered future displacements are encoded within the track embedding, while temporally indexed action tokens provide timestep-specific interactions through attention.

c) *Noise-dependent action grounding*: To obtain spatial anchors for action tokens, we combine the current end-effector position with a position derived from the noisy action at the current diffusion step. Let $\mathbf{g}_a$ denote the current world-frame end-effector position of arm a , and let $\tilde{\mathbf{a}}_{n,a}^k$ denote its

normalized noisy action at future step h and diffusion step k . We define

$$\mathbf{r}_{h,a}^k = \Phi_a(\tilde{\mathbf{a}}_{h,a}^k), \quad (21)$$

where Φ_a applies the inverse action normalization and extracts the absolute end-effector position of arm a . Both arms' positions are expressed in the same world frame. The spatial anchor is

$$\mathbf{x}_{h,a}^k = (1 - \rho_k)\mathbf{g}_a + \rho_k\mathbf{r}_{h,a}^k, \quad \rho_k = 1 - \frac{k}{K}, \quad (22)$$

where $k \in \{0, \dots, K - 1\}$ denotes the scheduler timestep on the training diffusion grid, rather than the DDIM iteration index. At inference, we use the scheduler-selected timesteps to compute ρ_k .

At high noise levels, the anchor remains close to the current end-effector position, reducing the influence of noisy action coordinates on rotary encoding. As denoising progresses, the anchor increasingly follows the position derived from the evolving action sequence. We recompute these anchors at every denoising step and use the same construction during training and inference. The anchors are computed from the current noisy actions, without using clean ground-truth future actions.